## DATASETS

Core traffic LSTM datasets

1. METR-LA (highway traffic sensors, LA)
    - Public, standard benchmark for spatiotemporal traffic forecasting (speed on sensor network).
    - Download (processed / CSV / npy):
        - Zenodo mirror (METR-LA + PEMS-Bay CSV): https://zenodo.org/records/5146275
        - Kaggle version: https://www.kaggle.com/datasets/annnnguyen/metr-la-dataset
    - Very good for LSTM/GNN speed prediction → can be used to estimate link travel times and feed a routing model, but does not include deliveries.
2. PEMS-BAY / Caltrans PeMS
    - Statewide 5-minute loop detector data (flow, speed, occupancy), used widely with LSTMs and DCRNN.
    - Ready-to-use PEMS-BAY:
        - Zenodo: https://zenodo.org/records/4263971
        - Kaggle: https://www.kaggle.com/datasets/scchuy/pemsbay
    - Raw PeMS portal (for custom districts/years): https://pems.dot.ca.gov/
    - Excellent for learning temporal congestion patterns; still needs a routing layer (OSM/graph) to turn forecasts into routes.
3. Helsinki traffic counters / mobility
    - Open traffic volume, speed, and vehicle-type data via API; updates every 5 minutes.
    - Traffic counters dataset: https://hri.fi/data/en_GB/dataset/liikennemaarat-helsingissa

- HSL (public transport, OD etc.): https://www.hsl.fi/en/hsl/open-data
- Good if you want a European city and combine car plus public transport or bikes in a sustainability narrative.

4. Uber Movement (historic travel times)
    - Aggregated average travel times and speeds between zones in many cities worldwide.
    - Access: https://movement.uber.com/ (if not reachable, see research mirror: https://data.transportationops.org/uber-movement)[11]
    - Useful for zone-to-zone travel time forecasting; less granular than raw GPS but aligns well with city-scale $CO_2$ analysis.

Mobility / taxi data

5. NYC TLC Taxi Trip Data
    - Fields: pickup/dropoff time & location, distance, fare, etc., from 2009 onward.
    - Official portal: https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page
    - AWS Open Data mirror: https://registry.opendata.aws/nyc-tlc-trip-records-pds/
    - Great for:
        - Travel time estimation between zones.
        - Temporal demand and congestion patterns for urban roads.
    - Not a route dataset by itself (each record is a single trip), but you can build an LSTM to predict trip time/ETA and use that inside a route-choice or assignment model.

Network / routing layer

6. OpenStreetMap + OSRM/GraphHopper

- OSM gives you the road graph; OSRM or GraphHopper provides routes and baseline travel times/distances.
- Typical workflow:
  - Download road network for a city (e.g., via Geofabrik) and load into OSRM/GraphHopper.
  - Use traffic datasets (METR-LA, PeMS, TLC, Helsinki, Uber Movement) to learn a correction factor with LSTM (time-of-day, weekday, weather) on OSRM's travel times.
- OSM description for HSL and routing: https://www.hsl.fi/en/hsl/open-data (section referencing OpenStreetMap use).

7. Google Maps / TomTom Traffic (paid APIs)
   - Very accurate real-time traffic flow and ETA, but not a public dataset; instead it's a query-based service with usage limits and fees.
   - Good for a production-style prototype, less suitable for fully reproducible academic work due to license and rate limits.

Logistics / last-mile specific datasets (you are missing these)

Your list is strong on traffic, but for "route optimization to reduce costs, carbon footprint, and delivery times" it helps to include data that actually contains stop sequences or planned vs. executed tours. Consider adding:

8. Last-mile delivery route deviations dataset (Mendeley)
   - Contains planned vs. actual delivery routes, timestamps, GPS traces; explicitly designed for studying route optimization and deviations.
   - Access: https://data.mendeley.com/datasets/kkwgfvmtxn
   - Very direct for benchmarking LSTM-based ETA prediction and route adherence, plus cost/$CO_2$ implications.
9. Amazon / e-commerce delivery datasets (Kaggle-style)
   - Various open datasets with order times, promised vs actual delivery time, hub locations, etc.

    - While not as standard as METR-LA, they give you real stop sequences and service times, which you can combine with traffic forecasts to compute operational cost and emissions.

10. Company-internal datasets (UPS ORION, DHL, etc.)

- As you noted, UPS ORION and DHL route datasets are not generally public; they appear only in case studies and literature.
- You can reference them conceptually in your paper but cannot rely on them as open benchmarks.

City of Chicago dataset status

Your "City of Chicago Traffic Tracker" idea is conceptually correct; Chicago has performance and congestion data, but it is less standardized for LSTM traffic forecasting than METR-LA/PeMS and may require significant cleaning and feature engineering. If you want a clean, widely cited benchmark, METR-LA and PEMS-BAY are safer.

What to pick for your project

For an M.Tech portfolio project that clearly shows LSTM + route optimization + sustainability:

- Best minimal combo (fully public, strong citations):
    - Use METR-LA or PEMS-BAY for LSTM traffic speed prediction.
    - Use OpenStreetMap + OSRM to convert predicted speeds into travel times and route costs.
    - Add NYC TLC or Uber Movement if you want an urban angle and larger scale.
- If you want explicit delivery routes:
    - Combine Last-mile route deviations dataset (Mendeley) with OSRM/OSM and maybe an external traffic dataset for calibration.
    - Frame the project around "LSTM-enhanced route planning for last-mile delivery: cost and $CO_2$ reduction".

If you tell the preferred city/region (e.g., US highways vs dense city vs Europe) and whether you want actual delivery stops or just generic traffic, a concrete 2-dataset combo can be recommended with a specific modeling pipeline.